Day 6 morning — Logistic regression

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Tuesday 17 November 2026 · Niels Jernes Vej 8A, room 0.02 (group work also in 0.01)

Yes/no outcomes · probability, odds and log-odds · priors on the log-odds scale · a design matrix in Stan · predicted probabilities and a grouped check

Today

08:45	**Lecture A**	Yes/no outcomes · logistic regression · priors on log-odds · predictions
09:45	**Exercise A**	Hit or miss: logistic model in Stan · predictions · check · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Simpson's paradox · the workflow · the final assignment
13:30	**Exercise B**	Does feedback help? A pilot study with a paradox · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

Follow-up from yesterday

From the Day 5 diaries and plenary

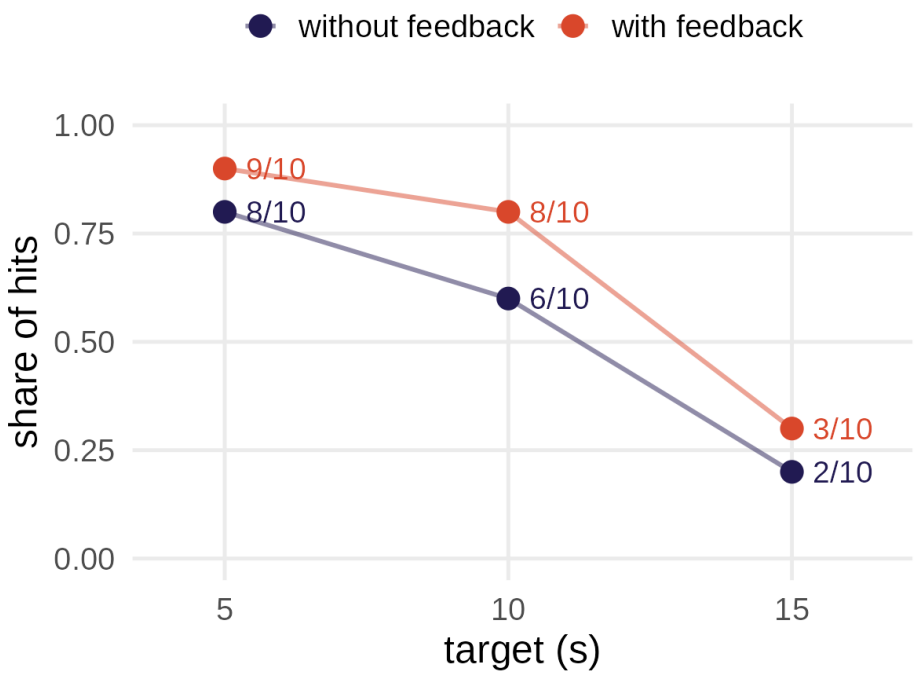
Fill in the evening before: the two or three points that came up most often under “not yet familiar”.

1. …
2. …
3. …

By the end of this lecture you can

1. **Model** a yes/no outcome with a probability that depends on predictors
2. **Move** between probability, odds and log-odds
3. **Set** priors on the log-odds scale and check them by simulation
4. **Fit** a logistic regression in Stan with a design matrix
5. **Report** odds ratios, predicted probabilities and a grouped check

Hit or miss?



One person, both Day 1 games: ten attempts at each target without feedback, then with feedback (illustrative numbers). A hit: her recorded time is within 0.5 s of the target.

Questions

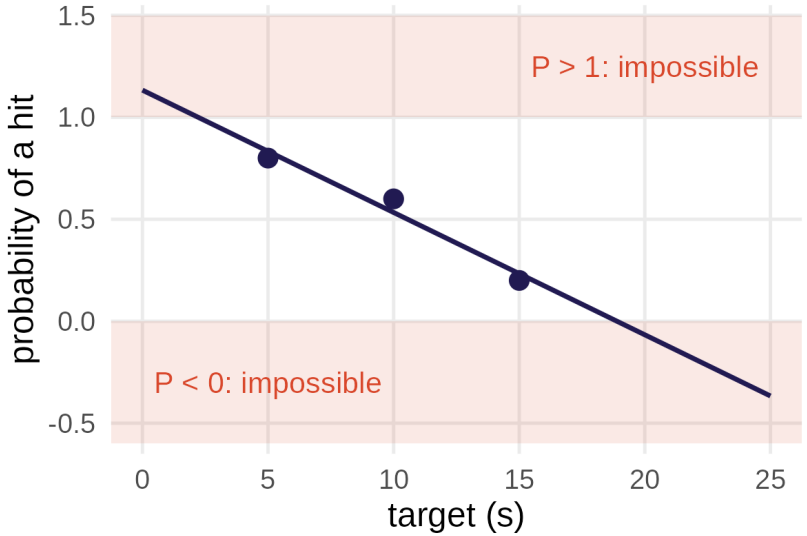
1. How does her chance of a hit fall with the target?
2. Does feedback help?
3. At 15 s with feedback: P(hit)? How many hits in her next 10 attempts?

The outcome is yes or no: we model a probability of yes.

A yes/no outcome

y_i ~ Bernoulli(p_i): $P(y_i = 1) = p_i$, $P(y_i = 0) = 1 - p_i$

- y_i = 1: attempt i is a hit; y_i = 0: a miss.
- Day 2: one p for every attempt, so the number of hits is binomial.
- Now p_i depends on the attempt: its target and whether she had feedback.
- We need a rule that turns predictors into a number between 0 and 1.
- A straight line for p does not stay there (right): through her three points it gives 1.13 at a 0-s target and −0.37 at 25 s.

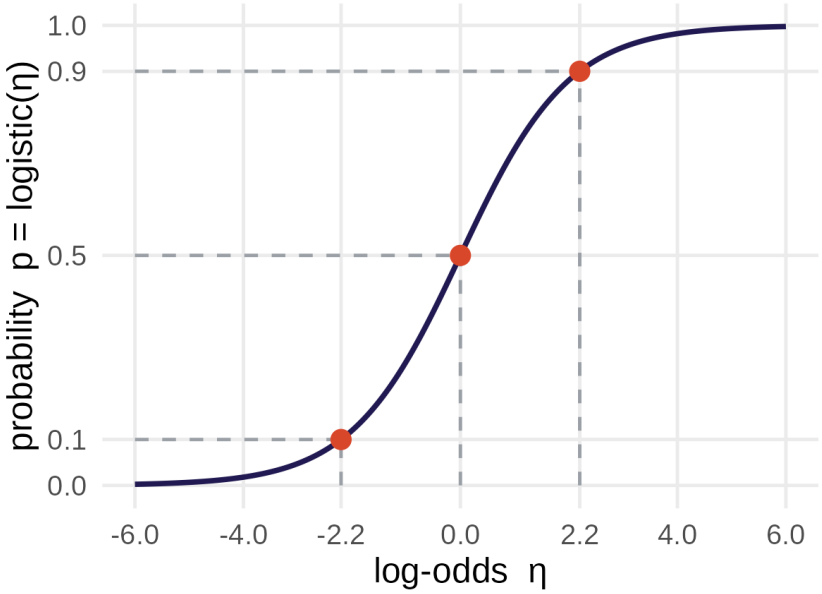


points: her hits without feedback
line: a straight line through them

Probability, odds and log-odds

probability p	odds p / (1 − p)	log-odds log(odds)
0.10	0.11	−2.2
0.25	0.33	−1.1
0.50	1	0
0.75	3	1.1
0.90	9	2.2

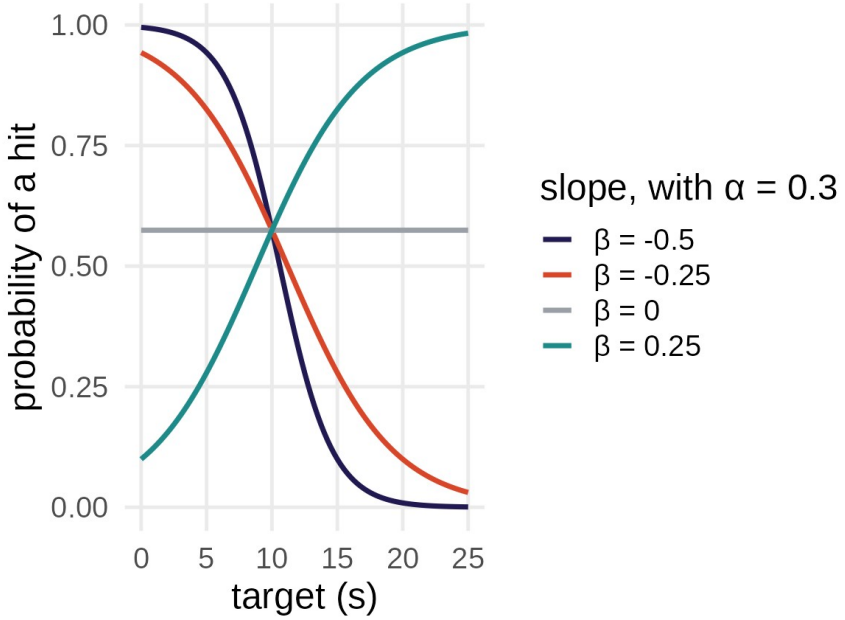
- logit(p) = log(p / (1 − p)) can be any number.
- Back again: p = logistic(η) = 1 / (1 + e^−η), always between 0 and 1.
- R: qlogis(p) and plogis(eta). Stan: logit() and inv_logit().



Logistic regression generates yes/no data

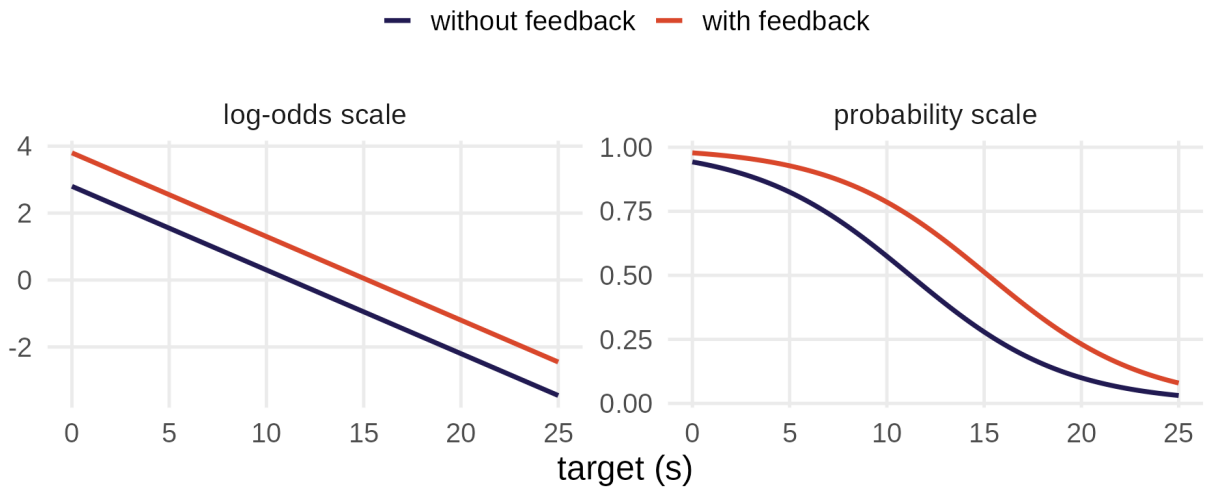
$$y_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \alpha + \beta \cdot x_i, \quad x_i = \text{target}_i - 10$$

- To simulate one attempt: η = α + β · x, then p = logistic(η), then y = 1 with probability p.
- α: log-odds of a hit at a 10-s target.
- β: change in log-odds per extra second; β < 0: hits get rarer at long targets.
- Every curve stays between 0 and 1. It is steepest at p = 0.5 and flat near 0 and 1.



A second predictor: feedback

$$\text{logit}(p_i) = \alpha + \beta_1 \cdot x_i + \beta_2 \cdot f_i, \quad f_i = 1 \text{ with feedback, 0 without}$$



$\alpha = 0.3$, $\beta_1 = -0.25$ per second, $\beta_2 = 1.0$ for feedback

- α: log-odds of a hit at 10 s without feedback. β_2: the shift with feedback.
- On the log-odds scale the shift is the same at every target (left); in probability it is not (right).

Reading the coefficients: odds ratios

Odds ratio

$\exp(\beta_2)$ = odds of a hit with feedback ÷ odds without, at the same target.

$\exp(\beta_1)$ = factor by which the odds change per extra second; per 5 s: $\exp(5\ \beta_1)$.

Probability

The change in p depends on where you start: largest near p = 0.5, small near 0 or 1. Near p = 0.5, one unit of x changes p by at most about β / 4.

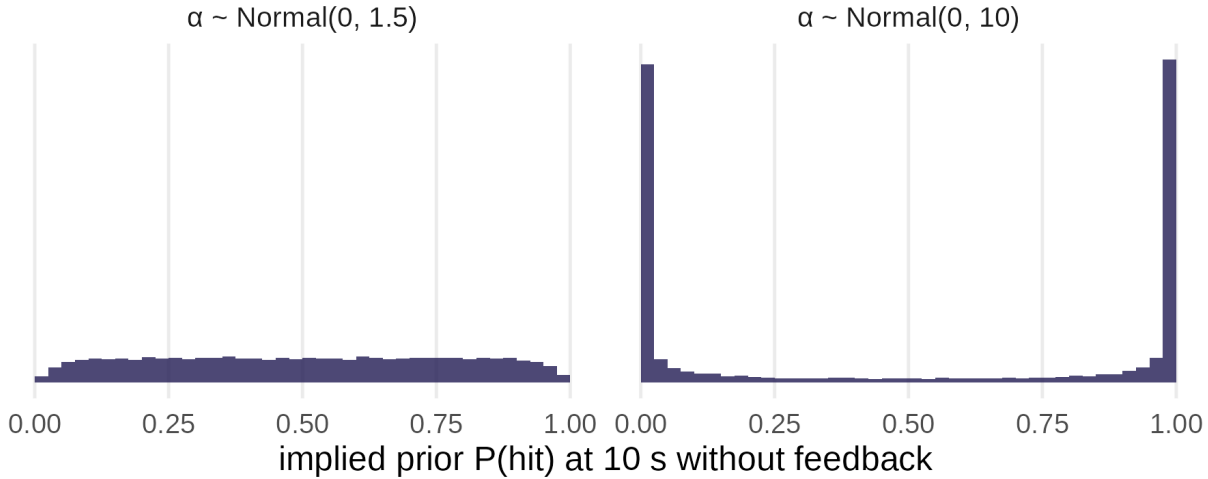
Example: $\alpha = 0.3$, $\beta_1 = -0.25$, $\beta_2 = 1.0$

Feedback: odds ratio $e^1 = 2.7$. At 10 s, P(hit) rises from logistic(0.3) = 0.57 to logistic(1.3) = 0.79.

Target: odds ratio e^−0.25 = 0.78 per second, e^−1.25 = 0.29 per 5 s.

Report effects as odds ratios — or, clearer, as probabilities for cases that matter.

Priors on the log-odds scale



- Push prior draws of α through logistic() and look at P(hit).
- Normal(0, 1.5): every P(hit) about equally plausible.
- Normal(0, 10): 65% of draws give P(hit) below 0.01 or above 0.99.

A "vague" prior on the log-odds is a strong prior on the probability (as on Day 3).

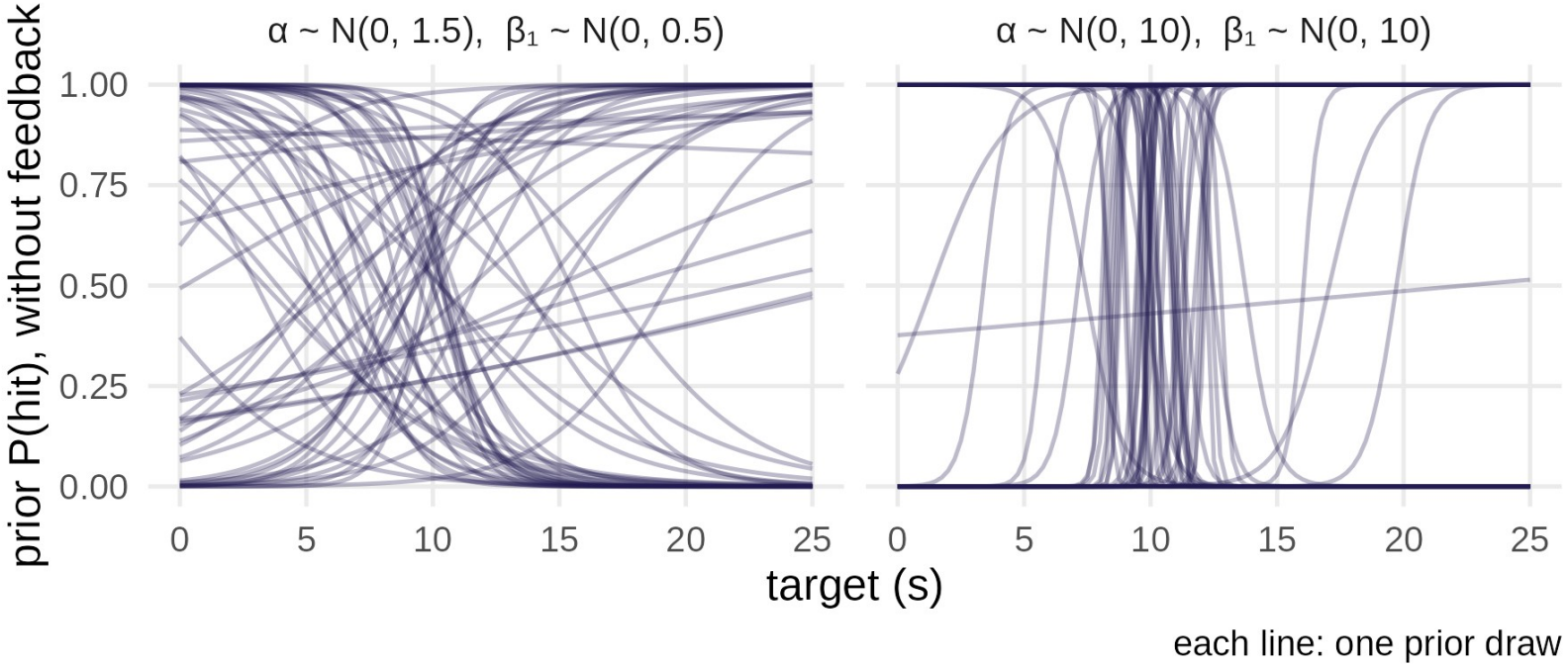
Priors for this model

parameter	prior	what it says
α	Normal(0, 1.5)	at 10 s without feedback, any P(hit) is plausible
β_1 (per second)	Normal(0, 0.5)	odds ratio per 5 s mostly between 0.08 and 12 (±1 SD)
β_2 (feedback)	Normal(0, 1)	odds ratio with feedback mostly between 0.37 and 2.7 (±1 SD)

Different predictors live on different scales, so each coefficient gets its own prior SD. Neither prior on a slope prefers helping or hurting: both are centred at 0.

Then simulate: what do these priors say about P(hit) at each target?

Prior predictive: curves of P(hit)



Left: any shape is possible, but gently. Right: P(hit) jumps from 0 to 1 within a second.

The design matrix

attempt	target	feedback	x = target − 10	f
1	5 s	no	−5	0
11	10 s	no	0	0
31	5 s	yes	−5	1
60	15 s	yes	5	1

X: one row per attempt, one column per predictor

$$\text{logit}(p_i) = \alpha + (\text{row } i \text{ of } X) \cdot \beta = \alpha + \beta_1 x_i + \beta_2 f_i$$

- More predictors: more columns. The Stan program stays the same.
- A yes/no predictor becomes a 0/1 column.
- In R: X <- cbind(x = d$target_seconds - 10, feedback = d$feedback)

The logistic model in Stan

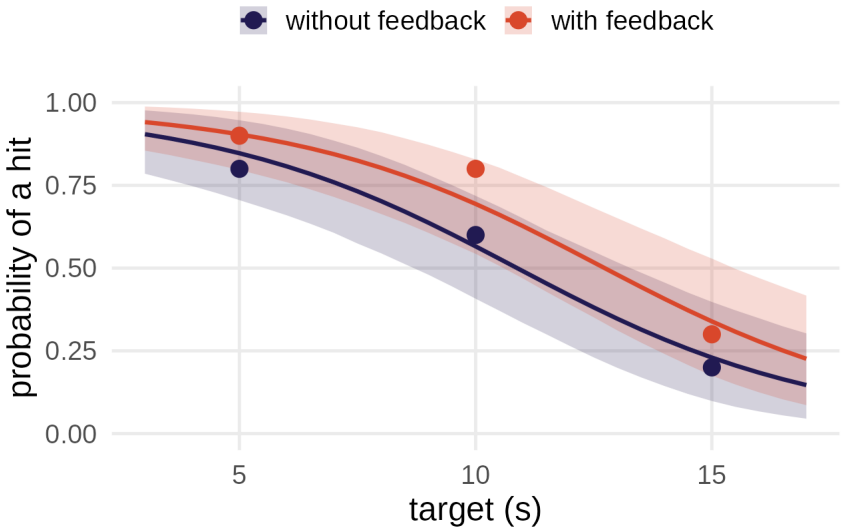
```
data {
  int<lower=1> N;  int<lower=1> K;
  matrix[N, K] X;                    // design matrix
  array[N] int<lower=0, upper=1> y;  // 1 = hit
  vector<lower=0>[K] prior_sd;
  int<lower=0> M;  matrix[M, K] X_new;
}
parameters {
  real alpha;
  vector[K] beta;
}
model {
  alpha ~ normal(0, 1.5);
  beta ~ normal(0, prior_sd);
  y ~ bernoulli_logit(alpha + X * beta);
}
generated quantities {
  vector[M] p_new = inv_logit(alpha + X_new * beta);
  array[N] int y_rep = bernoulli_logit_rng(alpha + X * beta);
}
```

- bernoulli_logit takes the log-odds directly: no inv_logit needed in the likelihood.
- prior_sd: one prior SD per column, passed as data.
- p_new: P(hit) for cases we choose, one row of X_new each.
- y_rep: a replicated dataset, for the check.

The fit

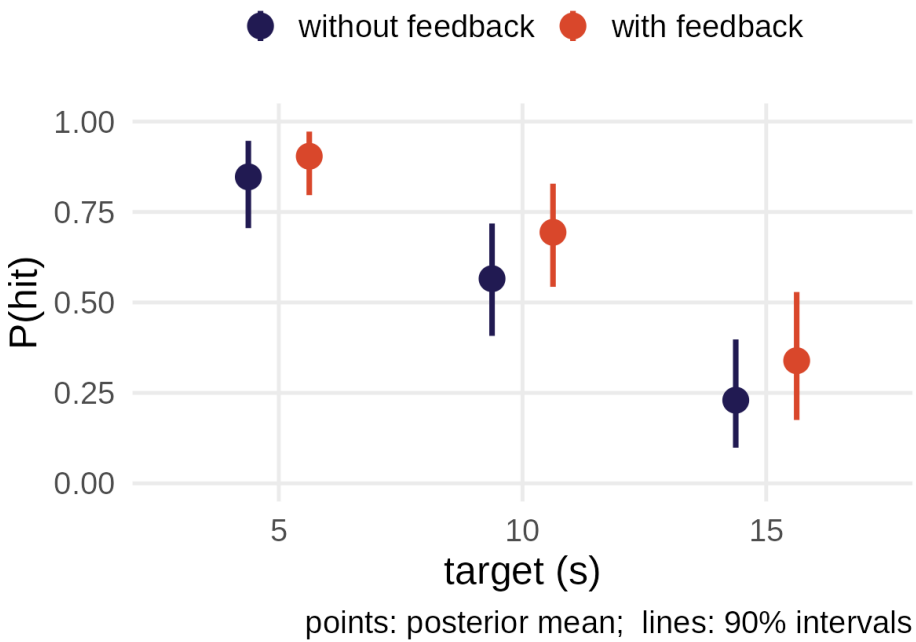
	mean	90% interval
α	0.27	−0.37 – 0.94
β_1 (per second)	−0.31	−0.46 – −0.18
β_2 (feedback)	0.58	−0.28 – 1.46

- No divergences, R-hat 1.00, ESS above 2,000.
- Odds ratio per extra second: 0.63 – 0.84. Each second lowers the odds of a hit by about a quarter.
- Odds ratio for feedback: 0.75 – 4.3; $P(\beta_2 > 0 \mid y) = 0.86$. Feedback probably helps, but ten attempts per target cannot pin it down.



lines: posterior mean; bands: 90% intervals; points: her share of hits

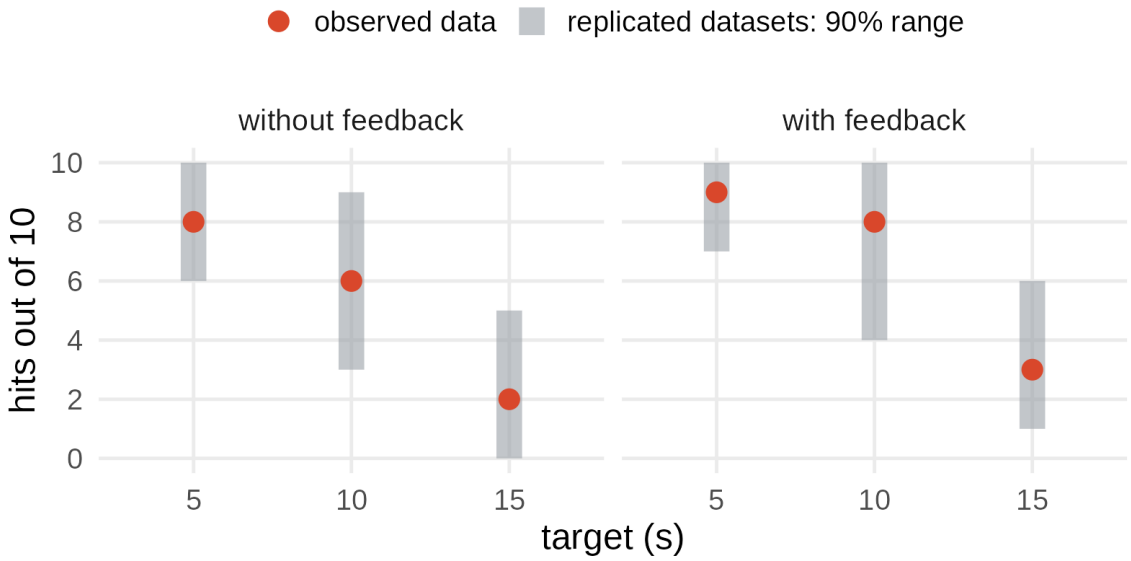
Predicted probabilities for chosen cases



- One row of X_new per case: target 5, 10, 15 s, without and with feedback.
- At 15 s with feedback: P(hit) = 0.34 (90%: 0.18 – 0.53).
- Her next 10 attempts there: for each draw, a binomial count with that draw's p.
- Result: 1 – 7 hits (90%); P(5 or more) = 0.27.

```
y_next <- rbinom(nrow(draws), 10,
draws$`p_new[6]`)
```

A predictive check for yes/no data



- A single 0 or 1 can never look unusual: check grouped counts.
- Here: hits out of 10 for each target and condition, from y_rep.
- All six observed counts lie inside the replicated ranges: no misfit in this statistic.

Group yes/no data to check them: hits per group, observed versus replicated.

Worked example: reading a logistic regression

Question 1

A report says: “$\beta_2 = 0.58$, so feedback raises the probability of a hit by 0.58.” What is wrong? What should be reported?

Question 2

With $\alpha = 0.3$ and $\beta_1 = -0.25$, what is P(hit) at a 20-s target without feedback? Compute it by hand.

Question 3

Why is the prior $\alpha \sim \text{Normal}(0, 10)$ not “uninformative” here?

Worked example: answers

Question 1

0.58 is a change in log-odds, not in probability. The odds ratio is e^0.58 ≈ 1.8 (90%: 0.75 – 4.3). In probability the change depends on the target: about 0.57 → 0.69 at 10 s, 0.23 → 0.34 at 15 s. Report probabilities for cases, with intervals.

Question 2

$\eta = 0.3 - 0.25 \times (20 - 10) = -2.2$, so $p = 1 / (1 + e^{2.2}) = 0.10$.

But she never tried 20 s: this is an extrapolation.

Question 3

On the probability scale it puts 65% of its weight below 0.01 or above 0.99: it claims she almost always or almost never hits. Judge priors on the scale you can picture, by prior predictive simulation.

Exercise A · 09:45 – 11:10

- Hit or miss: one person's 60 attempts from the two Day 1 games.
- Priors on the log-odds scale: prior predictive curves.
- Fit the logistic model in Stan with a design matrix; odds ratios.
- Predict P(hit) for given cases, and hits in 10 new attempts.
- A grouped predictive check. Your group's question set: concept · computation · pitfall.

Everyone back at

11:10

peer review A

Take a break when it suits your group. Rotate roles: coder · checker · question writers · interviewer · reporter.

Peer review A · 11:10 – 11:55

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

What to take into lunch and Lecture B.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

```
sample(not_yet_drawn, 2)   # plenary draw
```